

Asymmetric development of strategic coordination across the life cycle *

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April 27, 2026

Abstract

Coordination problems are central to social and economic life and often involve multiple equilibria. Classical theories identify two fundamental principles for equilibrium selection: maximizing payoffs and minimizing strategic risk. Yet little is known about how these principles emerge over the life cycle. Using a large sample of more than 2,000 participants spanning childhood, adolescence, and early adulthood, we document a sharp developmental asymmetry between payoff-based and risk-based coordination. Payoff-based coordination is already prevalent at the earliest ages studied and remains remarkably stable across development, with similar trajectories across cognitive ability and gender groups. In contrast, risk-based coordination emerges gradually and increases steadily with age. This developmental gradient is highly heterogeneous: it is concentrated among high-ability individuals and largely driven by males, while low-ability individuals and females display considerably flatter trajectories. These findings show that coordination is not developmentally uniform and that sensitivity to strategic uncertainty is progressively acquired rather than automatically available, revealing that similar coordination outcomes can be supported by fundamentally different underlying mechanisms.

Keywords: Payoff-dominance, Risk dominance, Coordination, Age

JEL codes: C7, C99, D90

*We would like to thank Àlex Izquierdo and Daniel Montolio for their help on data collection and Cesar Mantilla for his comments. In addition, we are very grateful to all the schools that participated and were always proactive in collaborating. This research was supported by the Spanish Ministry of Economy and Competitiveness (PID2021-126892NB-I00 & PID2024-1566156629NB-100), Excelencia-Junta (PY-18-FR-0007) and Agencia Andaluza de Cooperación Internacional para el Desarrollo (AACID-0I008/2020). The entire experiment was approved by the Ethical Committee of Loyola University Andalusia.

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Significance Statement

Coordination underlies many fundamental social and economic interactions, yet little is known about how individuals acquire the principles that enable successful coordination. Using a large sample spanning childhood, adolescence, and adulthood, we uncover a sharp developmental asymmetry between two canonical coordination principles. Payoff-based coordination is already prevalent at the earliest ages studied and remains remarkably stable across development, with similar trajectories across cognitive ability and gender groups. In contrast, risk-based coordination develops gradually and increases steadily with age. This developmental gradient is highly heterogeneous: it is concentrated among high-ability individuals and largely driven by males, while low-ability individuals and females display considerably flatter trajectories. These findings reveal that similar coordination outcomes can be supported by different underlying mechanisms and that sensitivity to strategic uncertainty is progressively acquired rather than automatically available.

1 Introduction

Classical theories of coordination treat payoff dominance and risk dominance as alternative equilibrium selection principles available to fully formed agents (**schelling1960strategy**; Cooper et al., 1990; Harsanyi & Selten, 1988). When multiple equilibria exist, individuals may coordinate on the outcome that maximizes joint payoffs or on the option that minimizes potential losses under miscoordination. Yet this symmetry in theory leaves an open developmental question: do these principles emerge simultaneously, or do they follow distinct trajectories across the life cycle?

Coordination problems pervade social and economic life—from adopting technologies and social norms to organizing collective action (**schelling1960strategy**; Camerer, 2003). In such environments, success does not only depend on incentives but also on shared expectations about others’ behavior. Canonical coordination models formalize payoff dominance and risk dominance as coexisting selection principles (Cooper et al., 1990; Harsanyi & Selten, 1988). While these principles are theoretically symmetric, empirical work has rarely isolated them, making it difficult to assess how individuals learn to rely on one or the other.

Understanding how coordination develops is particularly important given the growing literature on economic behavior in childhood and adolescence. Experimental evidence shows that preferences, beliefs, and decision-making abilities evolve substantially over this period (Brocas & Carrillo, 2018, 2021b; Glätzle-Rützler et al., 2025; Sher et al., 2014; Sutter et al., 2019). A large body of work documents systematic age patterns in risk taking (Harbaugh et al., 2002; Sutter et al., 2013), patience (Castillo et al., 2011; Sutter et al., 2013), social preferences (Almås et al., 2010; Fehr et al., 2008, 2013; Sutter & Kocher, 2007), and strategic reasoning (Brocas & Carrillo, 2020a, 2020c; Czermak et al., 2016), suggesting that adult economic behavior reflects a process of gradual development rather than a fully formed set of capabilities from early ages.

However, evidence on coordination itself remains limited and inconclusive. Recent experimental studies with children and adolescents find little or no robust age gradient in equilibrium selection in one-shot coordination games, despite clear sensitivity to incentives and beliefs (Glätzle-Rützler et al., 2025; Hermes & Schunk, 2022). The picture is more nuanced once interaction is repeated: Brocas and Carrillo (2021a) document that older adolescents are better able to converge on the efficient equilibrium in a repeated coordination setting, suggesting that learning may drive part of the developmental gap. Broader work on strategic reasoning likewise points to a stabilization of performance in normal-form interactions by mid-adolescence (Czermak et al., 2016). This raises a puzzle: if coordination behavior appears stable across development, does this reflect early mastery of coordination principles, or does aggregate behavior mask heterogeneity in the underlying reasoning processes?

We argue that payoff-based and risk-based coordination are not developmentally equivalent. Payoff-based coordination requires recognizing that higher rewards are preferable and expecting others to reason similarly—an intuition that may be available early in life. Risk-based coordination, by contrast, requires anticipating mistakes by others, reasoning about counterfactual outcomes, and responding to strategic uncertainty (Carlsson & Van Damme, 1993; Dal Bó et al., 2021; Heinemann et al., 2009). This type of reasoning is cognitively more demanding and may emerge gradually as strategic sophistication develops (Alaoui & Penta, 2016; Brocas

& Carrillo, 2018, 2020b, 2021b).

To test this idea, we study coordination behavior in a large sample of more than 2,000 participants spanning childhood, adolescence, and early adulthood. Participants play two coordination games designed to isolate payoff-based and risk-based principles. Crucially, we go beyond observed choices and elicit stated reasoning, allowing us to identify when coordination principles are deliberately applied rather than arising from behavior that is observationally equivalent but not grounded in strategic understanding (Costa-Gomes et al., 2001; Johnson et al., 2002).

We uncover a sharp developmental asymmetry. Payoff-based coordination is already prevalent at the earliest ages in our sample and remains remarkably stable across development. In contrast, risk-based coordination increases steadily with age, revealing a gradual acquisition of sensitivity to strategic uncertainty. This contrast points to a fundamental difference in underlying mechanisms: while payoff-based coordination appears to rely on an early and stable heuristic, risk-based coordination reflects a more demanding form of strategic reasoning that develops progressively over the life cycle. These findings reconcile the absence of strong age effects in aggregate coordination behavior with a structured developmental process in underlying reasoning. Rather than reflecting a uniform ability to coordinate, similar observed choices can arise from fundamentally different cognitive mechanisms.

2 Methods

2.1 Participants

The final analytical sample consists of 2,131 participants spanning late childhood, adolescence, and early adulthood. The data combine two studies that implement the same two-game coordination design and the same elicitation of post-choice justifications. Study 1 is a large lab-in-the-field classroom experiment conducted in Spanish schools, covering participants from late primary school to early adulthood. Study 2 is a laboratory experiment conducted at Universidad Loyola Andalucía with older participants, including high-school visitors and university students. Detailed information on recruitment, design features, incentive schemes, and sample composition is provided in the SI Appendix, sections A1 and A2.

2.2 Experimental design

Participants played two one-shot coordination games designed to isolate payoff-based and risk-based coordination principles. In both games, two players simultaneously chose between two actions. Payoffs were positive only when both players selected the same action, whereas miscoordination yielded lower or zero payoffs. The games were implemented using colored buttons rather than abstract labels, and the mapping between colors and actions was randomized across participants. Payoff magnitudes differed across studies: in Study 1, payoffs were scaled up by a factor of ten relative to Study 2. This rescaling does not affect the ordinal structure of the games or the equilibrium predictions. Table 1 reports the payoff matrices for the two games.

In the payoff-based coordination game, one equilibrium strictly Pareto-dominated the alternative, making payoff maximization the salient coordination principle. In the risk-based coordination game, both equilibria yielded identical payoffs under

successful coordination but differed in their robustness to unilateral deviations, giving rise to a unique risk-dominant equilibrium. Full experimental instructions and additional details are provided in the SI Appendix.

<i>Game 1 — Payoff-dominant coordination game</i>			<i>Game 2 — Risk-dominant coordination game</i>		
	Yellow (A)	Red (B)		White (A)	Blue (B)
Yellow (A)	1, 1	0, 0	White (A)	1, 1	0.5, 0
Red (B)	0, 0	0.5, 0.5	Blue (B)	0, 0.5	1, 1

Table 1: Payoff matrices of the coordination games. In Game 1, (A, A) is the payoff-dominant equilibrium. In Game 2, both equilibria yield identical payoffs under coordination, but (A, A) is the unique risk-dominant equilibrium.

2.3 Actions and stated reasoning

After choosing an action in each game, participants were asked to state the main reason for their decision by selecting one option from a predefined list presented in random order. In Game 1, the options were: (a) “Because I earned more”, (b) “Because I earned less”, (c) “Because I liked the color” and (d) “I chose one randomly”. In Game 2, the corresponding options were: (a) “Because it was less risky”, (b) “Because it was more risky”, (c) “Because I liked the color” and (d) “I chose one randomly”. The exact wording of the response options is reported in the SI Appendix, section A1.3.

We define deliberate payoff-based coordination (PayD) as selecting the payoff-dominant action (A) and reporting the corresponding justification (“Because I earned more”). Similarly, we define deliberate risk-based coordination (RiskD) as selecting the risk-dominant action (A) and reporting the corresponding justification (“Because it was less risky”).

2.4 Cognitive ability

Cognitive ability was measured using a composite index constructed as the sum of three components: the Cognitive Reflection Test (CRT; 0–3), financial abilities (0–3), and probabilistic reasoning ability (0–3), yielding a total score ranging from 0 to 9 (see SI Appendix, section A1.4, for tasks details). For university participants and students of visiting the university in the open day (Study 2), only CRT scores were available; to preserve comparability with the 0–9 scale used in the rest of the sample, these scores were rescaled by multiplying them by three.

To isolate within-age differences in cognitive performance from general developmental and sample-composition differences, we residualized the cognitive index with respect to age, gender, repeater status, class size, school fixed effects, and educational-stage fixed effects. We then standardized the resulting residuals to mean zero and unit variance. This standardized residualized measure, denoted $CogAb(r)$, is used in the main analyses.

2.5 Statistical analysis

We begin by presenting non-parametric descriptive evidence by age, cognitive ability, and gender. For the main analysis, we estimate linear probability models in which the dependent variable is either deliberate payoff-based coordination (PayD) or deliberate risk-based coordination (RiskD). The baseline specification includes age, the standardized residualized measure of cognitive ability ($CogAb(r)$), gender, and a set of controls, including repeater status, class size, incentive condition, and school fixed effects.

To examine heterogeneity in developmental trajectories, we augment the baseline specification with interaction terms between age and cognitive ability ($Age \times CogAb(r)$) and between age and gender ($Age \times Female$). To capture the non-linear relationship between cognitive ability and RiskD documented in the non-parametric analysis (SI Appendix, Figure A2), we also estimate specifications including a quadratic term in $CogAb(r)$.

All models are estimated using ordinary least squares with robust standard errors. These specifications correspond to the models reported in Table 2. Additional analyses are reported in the Supporting Information. Tables A9–A11 present the same specifications on progressively expanded subsamples excluding older participants. Similarly, Table A12 tests a triple interaction between age, gender, and cognitive ability, while Table A13 replicates the main results using logit models and reports average marginal effects. Finally, Table A14 replicates Table 2 replacing the CRT $\times 3$ rescaling for Study 2 participants with multiple imputation by chained equations (MICE-PMM).

3 Results

We begin by documenting the developmental pattern of payoff-based coordination. Figure 1 shows the proportion of participants choosing the payoff-dominant action (P) and deliberately applying payoff-based reasoning (PayD) across age. Overall, 76.0% of participants select the payoff-dominant action, and 60.7% meet the stricter criterion for PayD—substantially above the corresponding random benchmarks (0.50 and 0.125, respectively).

Strikingly, payoff-based coordination exhibits no meaningful developmental gradient. The correlation between age and PayD is small and statistically insignificant ($r = 0.032$, $p = 0.136$), and remains negligible after excluding the idiosyncratic dip at age 13 ($r = 0.018$, $p = 0.441$). Thus, deliberate payoff-based coordination is already prevalent at the earliest ages in our sample and remains remarkably stable throughout adolescence and into early adulthood.

This stability extends across subgroups. As shown in Figure 1B, high-ability individuals display higher levels of PayD than low-ability individuals (66.5% vs. 51.5%; $t = 7.011$, $p < 0.001$), replicating the well-established link between cognitive skill and strategic sophistication (Fe et al., 2022; Gill & Prowse, 2016; Johnson et al., 2002). Yet the age profile is flat within both groups ($r = 0.024$, $p = 0.380$ for high-ability; $r = 0.029$, $p = 0.412$ for low-ability), confirming that cognitive ability shifts the level of PayD but not its developmental trajectory. Similarly, Figure 1C reveals a modest gender difference in levels (males: 63.1% vs. females: 58.0%; $t = 2.411$, $p = 0.016$), yet no systematic age trend emerges for either group.

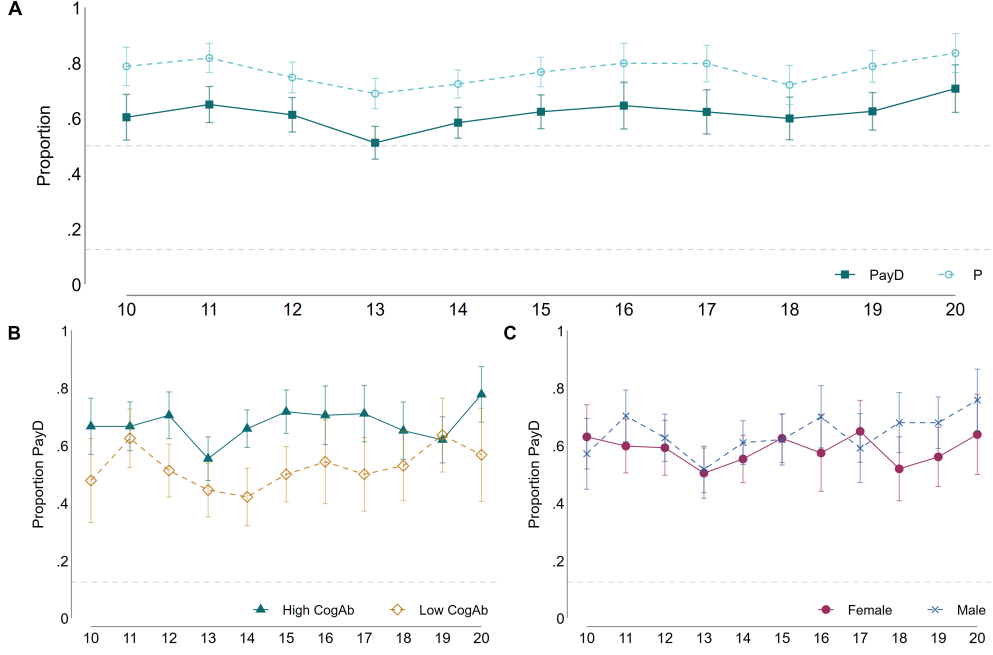


Figure 1: Payoff-based coordination across the life cycle. (A) Proportion of participants choosing the payoff-dominant action (*P*) and deliberately using payoff-based coordination (*PayD*) across age. Deliberate coordination is defined as selecting the payoff-dominant action and explicitly stating the corresponding reasoning. Payoff-based coordination is already prevalent at early ages and remains broadly stable throughout adolescence and into adulthood. (B) Deliberate payoff-based coordination by cognitive ability (high vs. low, split at the median). Higher cognitive ability is associated with higher overall levels of payoff-based coordination, but the age profile remains largely parallel across groups. (C) Deliberate payoff-based coordination by gender. No systematic gender differences or differential developmental trajectories are observed. Error bars indicate 95% confidence intervals. Dashed lines represent the random-choice benchmarks (0.50 for *P*; 0.125 for *PayD*).

(female: $r = -0.003$, $p = 0.929$; male: $r = 0.064$, $p = 0.032$), with the significant correlation for males disappearing once age 13 is excluded ($r = 0.049$, $p = 0.133$). Taken together, these patterns reveal that payoff-based coordination is already well established at the earliest ages in our sample and remains essentially flat across the life cycle. Cognitive ability and gender shift overall levels, but neither dimension meaningfully alters its developmental trajectory.

Figure 2 reveals a strikingly different developmental pattern for risk-based coordination. Figure 2A displays the proportion of participants choosing the risk-dominant action (*R*) and deliberately using risk-based coordination (*RiskD*) across age. Overall, 52.8% of participants chose the risk-dominant action and 29.1% met the stricter criterion for *RiskD*. While *RiskD* substantially exceeds the random-choice benchmark of 0.125 across all age groups, the rate of choosing *R* falls below the 0.50 benchmark among younger participants, rising above chance only from age 13 onwards. In sharp contrast to *PayD*, *RiskD* increases steadily with age—from 22.2% at age 10 to 39.6% at age 19—and the correlation with age is positive and highly significant ($r = 0.108$, $p < 0.001$). This contrast reveals a fundamental asymmetry: while payoff-based coordination appears to rely on a simple and early-available heuristic,

risk-based coordination reflects a more demanding form of strategic reasoning that develops progressively over the life cycle.

This developmental gradient is strongly shaped by cognitive ability. As shown in Figure 2B, high-ability individuals display substantially higher levels of RiskD than low-ability individuals (34.8% vs. 20.0%; $t = 7.351$, $p < 0.001$). More importantly, the timing of the developmental increase differs markedly across groups. Among high-ability individuals, the upward age trend emerges early and is already significant within the school-age range (ages 10–16: $r = 0.102$, $p = 0.002$), before flattening in the post-school years (ages 17–20: $r = 0.054$, $p = 0.286$), suggesting that the developmental gradient is largely completed by late adolescence. In contrast, low-ability individuals show no detectable age gradient during the school years ($r = 0.025$, $p = 0.549$), with a delayed increase emerging only when the full age range is considered ($r = 0.098$, $p = 0.005$). This timing asymmetry is further illustrated by Locally Weighted Scatterplot Smoothing (LOWESS) estimates reported in Figure A2 of the Supporting Information (SI) Appendix, which shows an earlier and steeper increase among high-ability individuals and a delayed acceleration among low-ability individuals. Cognitive ability thus plays a dual role: it shifts the overall level of RiskD and shapes the timing of the developmental gradient, with high-ability individuals showing an earlier and steeper trajectory than their low-ability peers.

Figure 2C reveals a pronounced gender asymmetry. Males exhibit a strong and significant increase in RiskD already within the school-age range (ages 10–16: $r = 0.110$, $p = 0.002$), which persists across the full sample ($r = 0.156$, $p < 0.001$). By contrast, females show no significant age trend at any stage of the life cycle (all $p > 0.10$). Overall levels also differ (males: 31.8% vs. females: 26.1%; $t = 2.900$, $p = 0.004$).

Taken together, the regression results confirm a sharp developmental asymmetry between the two coordination principles. Payoff-based coordination is already prevalent at early ages and remains essentially flat across the life cycle: cognitive ability shifts its level, but neither cognitive ability nor gender meaningfully alters its trajectory. By contrast, risk-based coordination increases robustly with age and does so in a highly heterogeneous manner. Its development is characterized by nonlinearities concentrated among high-ability individuals and by a pronounced gender asymmetry, with the developmental gradient largely driven by males.

Table 2 reports OLS estimates of deliberate coordination on age, cognitive ability, and gender, including school fixed effects and baseline controls. Columns 1–4 examine PayD and columns 5–8 examine RiskD.

The results confirm a sharp asymmetry between the deliberate use of payoff-based and risk-based coordination principles. For payoff-based coordination (columns 1–4), the age coefficient is small and only marginally significant across specifications ($\beta \approx 0.019$ – 0.021 , $p < 0.10$), indicating the absence of a meaningful developmental gradient. Cognitive ability is a strong predictor of PayD levels ($\beta \approx 0.067$ – 0.068 , $p < 0.01$). The interaction between age and cognitive ability is negative and marginally significant ($\beta = -0.006$, $p < 0.10$), consistent with a mild convergence pattern: high-ability individuals start from a higher baseline and exhibit slightly flatter age profiles, while low-ability individuals show marginally steeper ones, though neither group displays a meaningful developmental trend. By contrast, the interaction between age and gender is not significant, and the quadratic term in cognitive ability is small and insignificant, indicating broadly parallel and

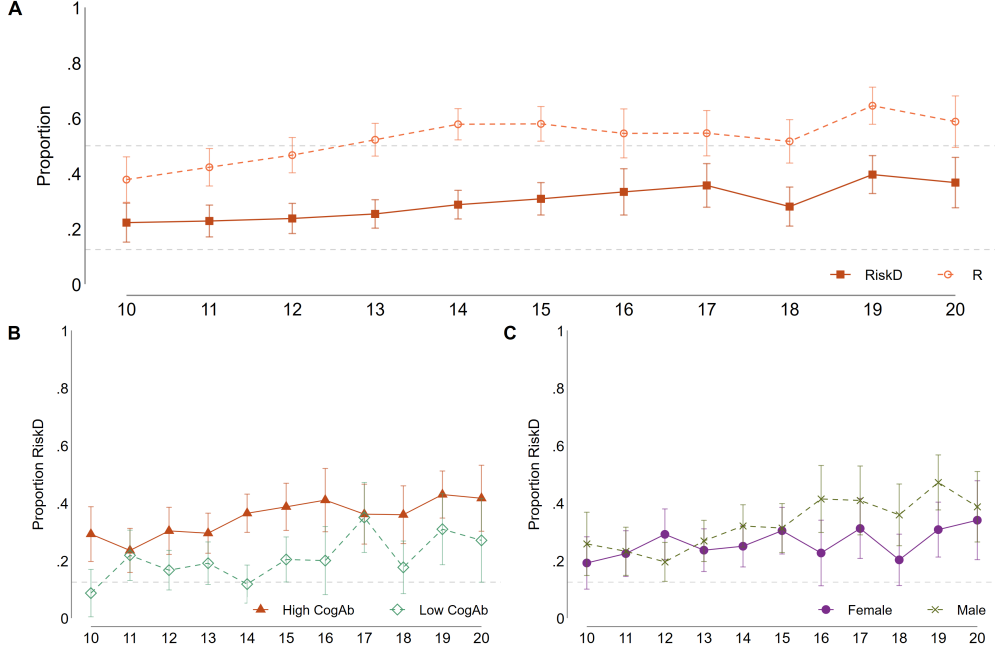


Figure 2: Risk-based coordination increases across development. (A) Proportion of participants choosing the risk-dominant action (R) and deliberately using risk-based coordination ($RiskD$) across age. Deliberate coordination is defined as selecting the risk-dominant action and explicitly stating the corresponding reasoning. Both measures increase monotonically from late childhood through adolescence and into adulthood. (B) Deliberate risk-based coordination by cognitive ability (high vs. low, split at the median within each age group). Higher cognitive ability is associated with greater overall use of risk-based coordination and a steeper developmental trajectory, while among low-ability individuals, the age trend is considerably flatter. (C) Deliberate risk-based coordination by gender. Males display a strong and significant increase in risk-based coordination with age, while females show a considerably flatter developmental trajectory. Error bars indicate 95% confidence intervals. Dashed lines represent the random-choice benchmarks (0.50 for R ; 0.125 for $RiskD$).

linear patterns across groups.

For deliberate risk-based coordination ($RiskD$; columns 5–8), the regression results confirm and extend the descriptive patterns. The age coefficient is positive, sizeable, and highly significant across all specifications ($\beta \approx 0.024\text{--}0.030$, $p < 0.01$), confirming a robust developmental increase in the deliberate use of risk-based coordination principles. Cognitive ability is also a strong positive predictor of $RiskD$ levels ($\beta \approx 0.071\text{--}0.072$, $p < 0.01$). Its interaction with age is small and not statistically significant ($\beta = -0.004$, $p > 0.10$), indicating that cognitive ability does not meaningfully alter the linear rate of increase in $RiskD$. Instead, consistent with Figure 2B and the non-parametric evidence in Figure A2 of the SI, differences across ability groups are better characterized by shifts in the timing and non-linearity of the developmental process, with high-ability individuals exhibiting earlier gains. This non-linearity is confirmed by the quadratic specification: the coefficient on $CogAb(r)^2$ is positive and significant ($\beta = 0.019$, $p < 0.05$), indicating that increases in $RiskD$ are concentrated among individuals with above-average cognitive ability. Finally, the interaction between age and gender is negative and statistically significant ($\beta = -0.012$, $p < 0.10$), indicating that the developmental increase in

RiskD is significantly weaker for females. This result aligns with Figure 2C and implies that the observed developmental gradient in deliberate risk-based coordination is largely driven by males.

Additional analyses reported in section C of the SI Appendix confirm the robustness of these findings. Tables A9–A11 replicate the main specifications on alternative samples—excluding university participants—and show that the age gradient in deliberate risk-based coordination is concentrated in the transition beyond compulsory secondary education, while payoff-based coordination remains stable throughout. Table A12 shows that the triple interaction between age, gender, and cognitive ability is negligible and not statistically significant for either outcome, indicating that the gender asymmetry in RiskD does not vary with cognitive ability. Table A13 reports logit specifications with average marginal effects, yielding qualitatively identical results. Finally, Table A14 replicates Table 1 replacing the CRT $\times 3$ rescaling for university participants with multiple imputation by chained equations (MICE-PMM, $M = 20$), yielding the same results.

Taken together, the regression results reveal a sharp developmental asymmetry between the deliberate use of payoff-based and risk-based coordination principles. Payoff-based coordination is already well established at the earliest ages and remains essentially flat across the life cycle: cognitive ability shifts overall levels, but neither cognitive ability nor gender meaningfully alters its developmental trajectory. Risk-based coordination, by contrast, increases robustly with age and does so in a highly heterogeneous manner. Its development is characterized by non-linearities concentrated among high-ability individuals and by a pronounced gender asymmetry, with the developmental gradient largely driven by males.

	(1) PayD	(2) PayD	(3) PayD	(4) PayD	(5) RiskD	(6) RiskD	(7) RiskD	(8) RiskD
Age	0.019* (0.010)	0.019* (0.010)	0.021** (0.010)	0.019* (0.010)	0.024*** (0.009)	0.024*** (0.009)	0.030*** (0.009)	0.024*** (0.009)
CogAb(r)	0.068*** (0.010)	0.162*** (0.054)	0.067*** (0.010)	0.070*** (0.011)	0.072*** (0.010)	0.138*** (0.050)	0.071*** (0.010)	0.079*** (0.010)
Female	-0.046** (0.021)	-0.048** (0.021)	0.031 (0.108)	-0.045** (0.021)	-0.053*** (0.019)	-0.054*** (0.019)	0.129 (0.099)	-0.049** (0.019)
Age $\times$ CogAb(r)		-0.006* (0.003)				-0.004 (0.003)		
Age $\times$ Female			-0.005 (0.007)				-0.012* (0.007)	
CogAb(r) ²				0.007 (0.008)				0.019** (0.007)
Constant	0.581*** (0.149)	0.584*** (0.149)	0.545*** (0.158)	0.575*** (0.150)	0.128 (0.139)	0.130 (0.139)	0.043 (0.146)	0.112 (0.139)
Observations	2,131	2,131	2,131	2,131	2,118	2,118	2,118	2,118
R-squared	0.078	0.080	0.079	0.079	0.065	0.066	0.066	0.068
CogAbB2	No	No	No	Yes	No	No	No	Yes
BIC	3034.3	3038.8	3041.5	3041.1	2739.2	2745.0	2743.4	2739.5
School FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Table 2: OLS regressions of deliberate coordination on age, cognitive ability, and gender. *The dependent variable is deliberate payoff-based coordination (PayD; columns 1–4) or deliberate risk-based coordination (RiskD; columns 5–8). Age is measured in years. CogAb(r) denotes age-residualized cognitive ability. Columns 2 and 6 include an interaction between age and CogAb(r) to test whether the developmental trajectory differs across cognitive ability groups. Columns 3 and 7 include an interaction between age and gender to test whether the developmental trajectory differs across genders. Columns 4 and 8 add a quadratic term CogAb(r)² to account for the non-linear relationship between cognitive ability and coordination documented in Figure S3. All specifications include school fixed effects and baseline controls (repeater status, class size, and monetary incentives according to each study). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

4 Discussion

This paper provides evidence that fundamental principles of coordination follow distinct developmental trajectories. Using a large sample spanning childhood, adolescence, and adulthood, we document a sharp asymmetry between payoff-based and risk-based coordination. Payoff-based coordination is already prevalent at early ages and remains broadly stable across development. In contrast, risk-based coordination increases steadily with age, even after accounting for cognitive ability. These findings indicate that strategic sophistication in coordination is not automatically available once cognitive skills are in place, but instead follows a structured developmental path, consistent with evidence on the gradual development of strategic reasoning Brocas and Carrillo, 2018, 2020b, 2021b.

A central contribution of our approach is to distinguish deliberate coordination from behavior that is observationally equivalent but not grounded in explicit reasoning. In simple 2×2 environments, selecting an equilibrium action may occur by chance. By eliciting both choices and stated reasoning, we identify when individuals deliberately rely on payoff or risk considerations. This distinction is particularly important in a developmental context, as it allows us to separate early intuitive coordination from the gradual acquisition of explicit strategic principles.

Our results have direct implications for canonical coordination models such as the

Stag Hunt. While payoff dominance and risk dominance are typically treated as alternative equilibrium selection principles available to fully formed agents (Schelling 1960; Cooper et al., 1990; Harsanyi & Selten, 1988), our findings show that they are not developmentally equivalent. Payoff-based coordination appears to rely on relatively simple reward-based reasoning that is accessible early in life. Risk-based coordination, by contrast, requires anticipating potential miscoordination and reasoning about others’ possible mistakes, which is closely related to the development of strategic reasoning about others (Sher et al., 2014). Sensitivity to strategic uncertainty is therefore not immediate, but progressively acquired over the life cycle.

More broadly, our findings suggest that learning to coordinate is a cumulative process. While individuals quickly internalize that higher payoffs are desirable, developing strategies that hedge against others’ errors requires experience in environments characterized by strategic uncertainty. This perspective helps explain why coordination failures may persist even among individuals with similar cognitive abilities: what differs is not only the capacity to reason, but the ability to anticipate and respond to others’ mistakes. Strategic experience—rather than cognitive maturation alone—plays a central role in the emergence of risk-sensitive coordination, in line with evidence linking cognitive skills to strategic sophistication (Fe et al., 2022; Gill & Prowse, 2016). The pronounced gender asymmetry in this developmental gradient—with males showing a consistently steeper trajectory than females—further suggests that the acquisition of risk-sensitive strategic reasoning may be shaped by broader differences in how males and females respond to strategic uncertainty, a pattern consistent with documented gender differences in competitive behavior and risk attitudes (Croson & Gneezy, 2009; Niederle & Vesterlund, 2007).

Author Contributions

P.B.-G. designed research; **P.L.** conducted the experiments in schools and constructed the dataset; **D.J.** performed the data analysis; **P.B.-G.**, **D.J.**, and **P.L.** wrote the paper.

Competing Interests

The authors declare no competing interests.

Data and Code Availability

The data and replication code supporting the findings of this study are publicly available in the Zenodo repository (<https://doi.org/10.5281/zenodo.19610263>).

AI Disclosure Statement

The authors used a generative artificial intelligence tool to assist with language editing and text refinement of the manuscript. All scientific content, analyses, and interpretations were developed and verified by the authors.

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